Sahaj Bhadja

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Summary

Software quality engineer with 2+ years validating **application behavior, data, integrations, and user workflows**; experienced translating requirements into repeatable test cases, reproducing defects, comparing expected and actual results, documenting evidence, and verifying resolutions.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Created **repeatable data-validation scenarios** for 5,000+ monthly workflows, carefully comparing expected states against application outputs and documenting malformed data, integration errors, reconciliation differences, and recovery behavior.
- Investigated application defects using logs, explicit job states, and targeted test data, carefully verifying corrective changes across PostgreSQL and SQL Server workflows while maintaining 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Executed **functional and regression testing** across PDF, DOCX, image, and ZIP workflows, identifying subtle content, formatting, OCR, and data differences while recording evidence and reproducible defect steps.
- Clarified requirements with users and developers, tested resolved issues across React interfaces, Flask APIs, and persistent results, and confirmed existing workflows remained stable across 10,000+ daily requests.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Maintained **cross-platform QA workflows** across Windows, Linux, and macOS VMs, executing installation, integration, and unit tests while generating CSV reports, code-coverage summaries, failure logs, and repeatable documentation.
- Collaborated with Microsoft and GitHub stakeholders to review requirements, reproduce API and synchronization defects, evaluate test implications, verify fixes, and sustain 20+ automated tests through nightly Jenkins execution.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Performed **comprehensive production end-to-end application testing** across eight services and 2M+ records, validating authentication, business rules, permissions, billing, notifications, WebSocket updates, reporting, and Azure SQL data consistency.
- Reproduced and documented defects using Sentry, Azure Monitor, and Kibana evidence, then carefully retested Docker releases through Jenkins to confirm fixes and maintain 95%+ successful build validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Converted customer requirements into **functional coverage** for booking, pricing, coupon, payment, refund, and status workflows, validating calculations, eligibility rules, webhooks, and Firebase data across 50,000+ records.
- Mapped 12+ connected workflows and clearly documented expected behavior, data relationships, and downstream outputs, collaborating with clients and developers to clarify discrepancies and verify sub-200-millisecond application responses.

Skills

Quality Assurance: Functional testing, regression testing, integration testing, test cases, defect reproduction, retesting
Validation: Expected-versus-actual analysis, data validation, business rules, visual inspection, evidence, test reporting
Platforms: macOS, Windows, Linux, React.js, REST APIs, PostgreSQL, SQL Server, Azure SQL, Firebase
Tools: PyTest, Mockito, Selenium, Postman, Sentry, Azure Monitor, Kibana, Git, Jenkins, GitHub Actions
Collaboration: Requirements clarification, user workflows, defect communication, documentation, Agile/Scrum, continuous improvement

Projects

PiXMod Live | *Next.js, Image Processing, MongoDB*

- Built an **image-processing application** for segmentation, recoloring, restoration, search, and subscription workflows, processing 1,000+ edited assets with SAM, Stable Diffusion, responsive Next.js interfaces, and MongoDB metadata storage.

HMIS Live | *MERN, Microservices, OOP*

- Tested a **healthcare workflow application** spanning role-based access, patient-history retrieval, appointments, dashboards, notifications, REST integrations, and Sentry monitoring, halving record retrieval time while reducing administrative workload 30%.

CPU Scheduling Simulator Live | *C++, Functional Testing, React*

- Created and executed **functional test scenarios** for six scheduling algorithms, comparing expected and actual outputs across preemptive, non-preemptive, and edge cases with dynamic metrics over 1,000+ process combinations.